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Chemistry–A European Journal
Research Article
doi.org/10.1002/chem.202401214
Chemistry Europe
European Chemical Societies Publishing
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HPLC-Based Automated Synthesis and Purification of Carbohydrates

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Reported herein is a new HPLC-based automated synthesizer (HPLC–A) capable of a temperature-controlled synthesis and purification of carbohydrates. The developed platform allows to perform various protecting group manipulations as well as the synthesis of *O*- and *N*-glycosides. A fully automated synthesis and purification was showcased in application to different carbohydrate derivatives including glycosides, oligosaccharides, glycopeptides, glycolipids, and nucleosides.

Introduction

Carbohydrates have been known as building blocks of nature or *"essential molecules of life"*[1] but understanding of their roles as "molecules of death" due to their involvement in "*in every major disease*,"[2] came to the fore only in the recent decades. With improved understanding of various roles of carbohydrates, invention of new methods and technologies to generate both natural sugars and their mimetics have come forward as the key priority of glycosciences. Complex carbohydrates can be isolated from natural sources[3] or synthesized, either enzymatically[4] or chemically.[5] Still, availability of glycans remains scarce.

Glycosides are obtained by glycosylation reaction, and the conventional synthesis of elongated glycan sequences involves additional deprotection or leaving group introduction steps after each glycosylation. Typically, aqueous work-up and purification of all reaction intermediates is required for most transformations.[6] The glycosylation-deprotection steps are then reiterated, but these syntheses are laborious and lengthy, which leads to poor availability of glycans. A majority of expeditious strategies that streamline oligosaccharide synthesis are based on chemoselective, selective, or orthogonal activation of leaving groups.[7] Approaches that rely on orthogonality or selectivity of temporary protecting groups are also known.[8] Regardless of the strategy chosen, both conventional and expeditious manual chemical synthesis of glycans remains challenging, which makes it hard to implement in non-specialized labs to accelerate innovations in other areas of glycosciences.

Solid-phase synthesis of glycans on polymer supports also involves glycosylation and deprotection steps, but the reaction work-up and purification of intermediates are bypassed,[9] which creates a promising platform for automation. Seeberger and co-workers have been developing glycan synthesizer, which was initially based on a peptide synthesizer,[10] and subsequently evolved into "*the first fully automated solid-phase oligosaccharide synthesizer*" in 2012.[11] This synthesizer was then commercialized as Glyconeer 2.1,[12] and the next generation, Glyconeer 3.1 was introduced in 2023. Also in 2012, Demchenko and Stine reported HPLC-based automation (HPLC–A) of solid phase synthesis.[13] Very recently, Hurevich implemented some important methodological and operational improvements into reactions on solid phase.[14] Hsieh-Wilson and Huang reported an automated solid phase synthesis of a heparan sulfate library.[15]

Automated synthesis of oligosaccharides in solution has been developed by Takahashi,[16] Pohl,[17] Nokami,[18] and us.[19] Recently, Ye reported a very impressive synthesis of a 1,080-mer glycan.[20] While still under development, automated synthesis in solution complements efforts to automate enzymatic synthesis by Wong,[21] Chen,[22] Wang,[23] and Boons.[24] First efforts to automate building block synthesis have also been reported by Bennett and Pohl.[7c, 25] The availability of accessible, dependable, and easily transferable methods for carbohydrate synthesis is necessary for boosting practical applications and further discoveries in all areas of glycosciences. Reported herein are our first steps towards developing a universal platform both for the synthesis of carbohydrate building blocks and for glycosylation. These reactions were followed by chromatographic purification using a completely automated HPLC-based sequence of operations. It should be noted that the *in-line* purification circuit implemented herein is a new capability compared to other automated glycan synthesizers. This platform was applied to the synthesis of targets and linkages that have been strategically chosen to test various reaction conditions, monitoring capabilities, column loading, programming, etc.

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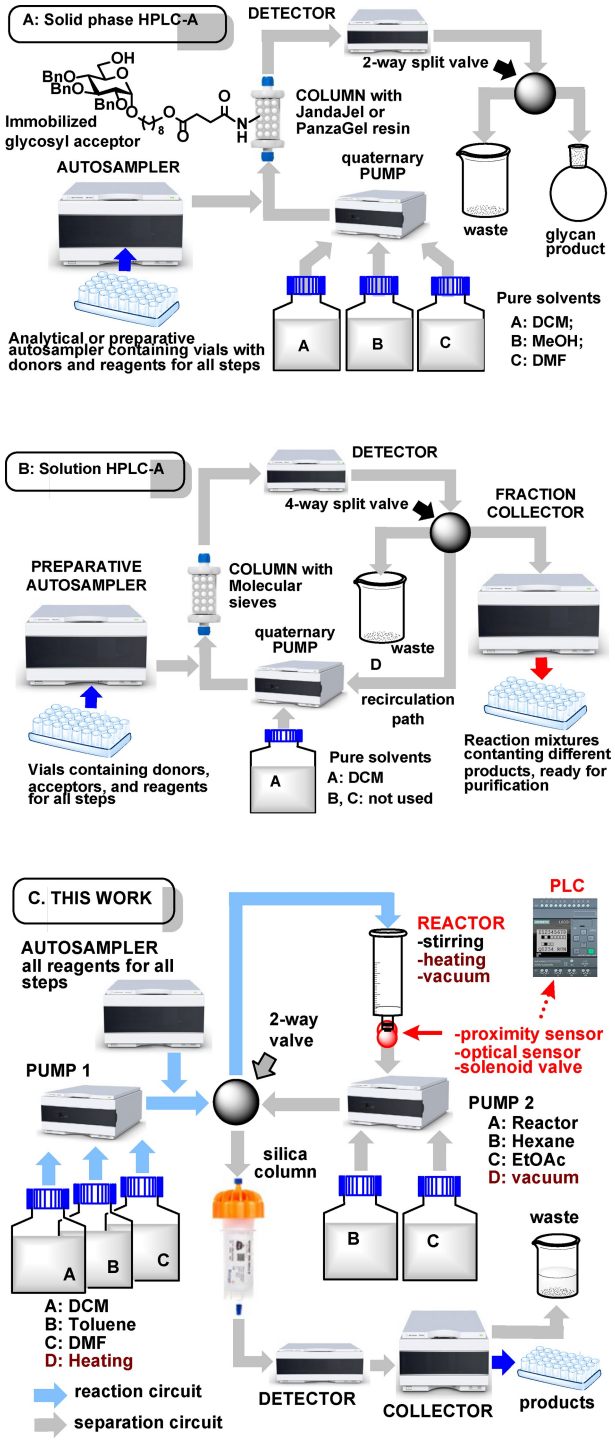
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Results and Discussion

In our original set-up, resin-immobilized acceptor was packed into an Omnifit column that was integrated into a HPLC system.[13] All reactants and reagents were delivered via intake lines of a HPLC pump. Glycosylations were performed by recirculating the premixed mixture containing the imidate donor and TMSOTf. The donor was equipped with a temporary Fmoc protecting group that was then selectively removed with a mild base. After the desired number of glycosylation and Fmoc deprotection iterations, the glycan was cleaved off from the resin. Although this set-up offered some operational ease and faster reaction times, it largely remained semi-manual.[13] Implementation of an autosampler for delivering the promoter for glycosylations further enhanced the utility of the HPLC–A.[26] However, this improved set-up remained semi-manual because shifting reaction vessels and switching between the reaction, product collection, and discharge modes required manual intervention by the operator. Subsequently, solid phase HPLC–A was supplemented with a 2-way split valve as a mode to achieve entirely automated process (Scheme 1A).[27] The sequence consisting of four glycosylation-deprotection cycles, followed by the off-resin cleavage and final product collection in a dedicated vessel was performed with the "press-of-a-button." All donors and reagents were delivered by the autosampler, and the pump supplied standard ACS-spec solvents.

Recently we introduced a new set-up for performing the synthesis of multiple glycan products with the single press-of-a-button.[19] To enable this capability, we switched to the solution phase synthesis and supplemented the system with a fraction collector and a four-way split valve. The implementation of the new valve allowed for completely automated switch between reaction/ recirculation, waste discharge, and the product collection modes (Scheme 1B). Pump line A was dedicated to delivering DCM solvent which was used for glycosylation and washing. Recirculation was performed via Pump line D, which was specifically dedicated for this purpose, whereas the autosampler delivered all building blocks and reagents. The column was repurposed to hold activated molecular sieves. The software was programmed to perform completely automated synthesis of up to six di- and tri-saccharides from randomly selected donor-acceptor combinations in 12 h. The individual reaction mixtures were then manually purified to obtain oligosaccharides in good to excellent yields.[19]

With the ultimate goal of developing scalable synthesis of building blocks, glycosylation, and purification in a fully automated regime, we interfaced our HPLC–A with the reactor-based synthesis. The modular character of HPLC gave us an advantage of implementing new components and attachments by using the plug-in approach and modulating reaction modes by computer programming. Our new set-up comprises two main circuits: the reaction circuit and the separation circuit (Scheme 1C). The reaction circuit (upstream operations) ➡ is equipped with a quaternary pump, an autosampler, a 25 mL glass-syringe-reactor, and it is operated by the standard HPLC software. The autosampler is programmed to deliver all necessary reagents to the reactor equipped with a stirrer. The reactor also has heating and cooling capabilities (*vide infra*), and the bottom outlet is controlled with a solenoid valve. The reaction mixture is then stirred for a given time, in most cases the same as for manual synthesis, which simplifies planning and converting of conventional manual reactions to automated processes. When necessary, the reaction can be quenched by delivering appropriate reagents via the autosampler.



Scheme 1. HPLC–A: solid phase (A), solution (B), this work (C).

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Post-reaction/separation circuit (downstream operations) is equipped with a quaternary pump, a disposable flash chromatography cartridge, a UV detector, and a fraction collector, all operated by the standard HPLC software. We have also added programmable logic controller (PLC) to control the crude reaction mixture transfer from the reactor to the chromatography column. PLC operates a capacitive proximity sensor installed at the bottom outlet of the reactor, an optical pathway sensor, and a solenoid valve. We engineered this circuit to control or interrupt the flow during the transfer by opening/closing the solenoid valve. After transferring the reaction mixture to the equilibrated chromatography column, the reactor is rinsed by small amounts of toluene (Pump 1), and this solution is also loaded onto the column. The sequence concludes with the chromatographic separation and the collection of fractions. The reaction and separation circuits are interfaced with a standard 2-way valve, which allows to use both pumps for both circuits. Whenever is necessary, the volume reduction, or unwanted volatile by-product removal, can be affected by engaging a mini vacuum pump via relay transduction of the electrical signal from line D of Pump 2.

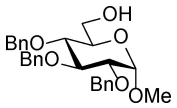
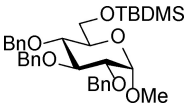
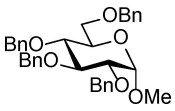
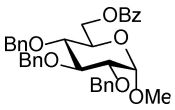
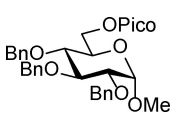
Full automation of a given reaction-purification is achieved by creating a target-specific operational sequence in one continuum. All sequences start with the washing cycle to clean all tubing in upstream circuits. The subsequent operation consists of commands to deliver all reagents and all solvents to the reactor. The common bulk ACS-grade solvents were delivered via respective intake lines of Pump 1. All reagents, reactants, and reaction-specific solvents and solutions were delivered with the autosampler. Concomitantly, Pump 2 was engaged to pass appropriate solvents to equilibrate the chromatography column. Upon completion of the reaction, subsequent sequence of operations included transfer of the reaction mixture from the reactor to the column, chromatographic separation, and automated collection of purified compounds enabled by the UV detector and fraction collector interface. All details of these operations and programming are included in the supporting information.

We began studying this new reaction-purification HPLC–A set-up by performing simple reactions involving the introduction of protecting groups commonly used in carbohydrate chemistry for the synthesis of building blocks. These reactions were all performed at rt, and the study was focused on refining all basic parameters and programming command sequences. Once the desired sequence of operations was written, the start button was pressed, and all operations including washing, reagent delivery, column equilibration, reaction, reaction mixture transfer, purification, and product collection were all performed in a completely automated manner.

We first investigated a silylation reaction. The autosampler was programmed to deliver solutions of TBDMSCl in DCM, 6–OH derivative **1**[28] in DCM, imidazole in DMF, and triethylamine (TEA) in DCM to achieve the reagent ratio listed in Table 1 (entry 1). Upon completion of the reaction (5 h), the reaction mixture was transferred to the column, and the purified product was collected by the automated fraction collector. The post-automation collection and concentration of the respective fractions afforded 6-*O*-TBDMS derivative **2**[29] in 81% yield. Benzylation, benzoylation, and picoloylation of compound **1** were then conducted in a similar manner. Besides benzylation, which resulted in a low yield of compound **3**[30] (36%, entry 2) due to the presence of a number of side products, other reactions provided the respective products **4**[31] and **5**[8c] in commendable yields (74–81%, entries 3–4).

With successful first attempts to engineer a suitable automated platform for the synthesis and purification of

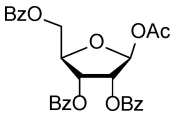
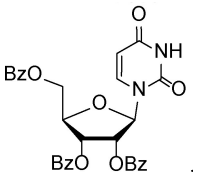
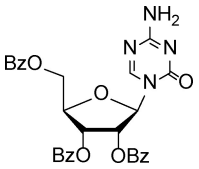
Table 1. Automated introduction of common protecting groups.

Entry	Substrate	Reaction mode and reagents	Time and temp	Solvents	Product	Yield
1	 **1**	**Automated** TBDMSCl (4.0 equiv) Imidazole (1.1 equiv) TEA (4.5 equiv)	5 h, rt	DCM, DMF	 **2**	81%
2	**1**	**Automated** Benzyl trichloroacetimidate (1.5 equiv) TfOH (0.5 equiv)	16 h, rt	DCM	 **3**	36%
3	**1**	**Automated** Benzoyl chloride (1.6 equiv) DMAP (0.2 equiv) TEA (1.5 equiv)	21 h, rt	DCM	 **4**	81%
4	**1**	**Automated** Picolinic acid (2.0 equiv) Ethylenediaminocarbodiimide (EDC, 6.0 equiv) DMAP (0.4 equiv)	16 h, rt	DCM	 **5**	74%

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Table 2. *N*-glycosylation in application to the synthesis of active pharmaceutical ingredients.

Entry	Substrate	Reaction mode and reagents	Time and temp	Solvents	Products	Yield
1	 **6**	**Manual** Step 1. Uracil (1.2 equiv) *N,O*-Bis(trimethylsilyl) trifluoroacetamide (BSTFA, 2.2 equiv) Step 2. **6**, TMSOTf (1.1 equiv) TfOH (0.5 equiv)	Step 1. 2 h, rt Step 2. 2 h, rt	DCM	 **7**	80%
2	**6**	**Automated** Step 1. Uracil (1.5 equiv) BSTFA (3.3 equiv) Step 2. 6, TMSOTf (1.2 equiv)	Step 1. 1 h, rt Step 2. 14 h, rt	DCM	**7**	90%
3	**6**	**Manual** Step 1. Azacytosine (1.0 equiv) BSTFA (2.4 equiv) Step 2. 6, TMSOTf (1.1 equiv) TfOH (0.5 equiv)	Step 1. 2 h, rt Step 2. 8 h, rt	DCM		86%
4	**6**	**Automated** Step 1. Azacytosine (1.2 equiv) BSTFA (2.3 equiv) Step 2. 6, TMSOTf (1.2 equiv) TfOH (1.0 equiv)	Step 1. 4 h, rt Step 2. 8 h, rt	DCM	**8**	91%

protected monosaccharides, we investigated this set-up in application to the synthesis of Active Pharmaceutical Ingredients (API). Among possible targets, we investigated syntheses of uridine and azacitidine ribosides via *N*-glycosylation of the respective bases (Table 2). The bases uracil and 5-azacytosine are poorly soluble in reaction solvents. In known manual syntheses the bases are added as solids, which was also adapted herein. To refine the reaction conditions, we performed manual synthesis first. Silylation of uracil with *N,O*-bis(trimethylsilyl) trifluoroacetamide followed by the reaction with acylated ribose **6** followed by *N*-glycosylation in the presence of TMSOTf and TfOH resulted in the formation of nucleoside **7**[32] in 80% yield (entry 1). The automated *N*-glycosylation was tested on a similar scale (100 mg of **6**). As in the previous approach, solid uracil was added manually, and the subsequent operation sequence including washing, reagents delivery, two reaction periods, column equilibration, reaction mixture transfer to the column, separation, and fraction collection was fully automated. As a result, nucleoside **7** was obtained in 90% yield (entry 2). Essentially the same approach was taken for the manual and automated syntheses of azacytidine nucleoside. As a result, product **8**[33] was obtained in 86–91% yield (entries 3–4). Interestingly in both of these examples, the automated approach consistently produced the desired compound in a 5–10% higher yield than the manual approach.

As the next step, we investigated reactions at different temperatures. To implement this capability, we first added the reactor heater along with the sensor depicted in Figure 1A. The heat rate was controlled via relay transduction of the electrical signal from the rewired line D of Pump 1. The relay transduces a low-power input current generated from a solvent gate of the



Figure 1. High-temperature set-up (A); low-temperature set-up (B).

HPLC pump as high-power switch output. For a full technical description of the components refer to the Supporting Information.

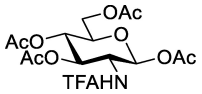
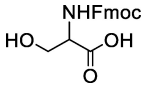
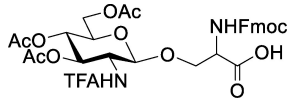
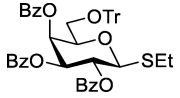
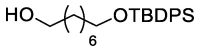
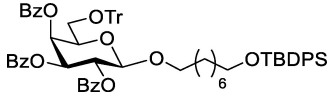
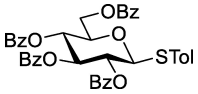
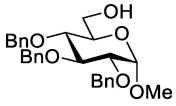
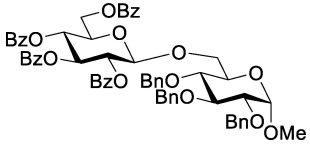
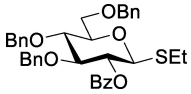
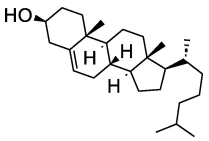
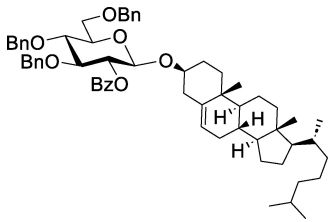
To test this capability, we chose glycosidation of glucosamine acetate donor **9**[34] with amino acid acceptor **10**, a reaction that works best at elevated temperatures (Table 3). With inspiration from previous reports,[35] we performed this reaction in the presence of catalytic $InBr_3$ in MeCN-dibromomethane as the reaction solvent at 80 °C. Conventional manual synthesis afforded the desired amino acid glycoside **11** in 79%

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Table 3. Temperature controlled *O*-glycosylation reactions.

Entry	Donor	Acceptor	Reaction mode and reagents	Time and temp	Solvents	Product	Yield
1			**Manual** $InBr_3$ (0.3 equiv)	8 h, 80 °C	MeCN, CH_2Br_2		79 %
	9 (5.0 equiv)	**10** (1.0 equiv)				**11**	
2	**9** (3.9 equiv)	**10** (1.0 equiv)	**Automated** $InBr_3$ (0.44 equiv)	8 h, 80 °C	MeCN, CH_2Br_2	**11**	78%[a]
3			**Manual** NIS (5.0 equiv) TfOH (0.3 equiv)	45 min, −20 °C	DCM, 1,4-dioxane		70%
	12 (2.0 equiv)	**13** (1.0 equiv)				**14**	
4	**12** (2.3 equiv)	**13** (1.0 equiv)	**Automated** NIS (5.6 equiv) TfOH (0.3 equiv)	1 h, −20 °C	DCM, 1,4-dioxane	**14**	80%
5			**Manual** NIS (2.0 equiv) TfOH (0.5 equiv)	1.5 h, −20 °C	DCM, MeCN		71%
	15 (1.0 equiv)	**1** (1.5 equiv)				**16**	
6	**15** (1.0 equiv)	**1** (1.5 equiv)	**Automated** NIS (2.0 equiv) TfOH (0.5 equiv)	1.5 h, −20 °C	DCM, MeCN	**16**	70%
7			**Manual** NIS (2.0 equiv) TfOH (0.4 equiv)	1 h, 0 °C	DCM, MeCN		88%
	17 (1.0 equiv)	**18** (2.0 equiv)				**19**	
8	**17** (1.0 equiv)	**18** (2.0 equiv)	**Automated** NIS (2.0 equiv) TfOH (0.4 equiv)	1 h, −5 °C	DCM, MeCN	**19**	84 %[b]

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Table 3. continued

Entry	Donor	Acceptor	Reaction mode and reagents	Time and temp	Solvents	Product	Yield
9	**17** (3.6 equiv)	**20** (1.0 equiv)	**Manual** NIS (5.2 equiv) TfOH (0.5 equiv)	1 h, −10 °C	DCM, MeCN	**21**	79%
10	**17** (3.5 equiv)	**20** (1.0 equiv)	**Automated** NIS (5.0 equiv) TfOH (0.6 equiv)	1.5 h, −10 °C	DCM, MeCN	**21**	76%[a]

[a] 90% purity. [b] 95% purity.

yield (Table 3, entry 1). After optimization of the reaction conditions, which comprised minor tweaking of the reagent ratios, we achieved essentially the same result using the automated approach. After fully automated reaction-separation sequence, glycoside **11** was isolated in 78% (90% purity, entry 2).

Subsequently, we investigated reactions at low temperatures, which are very common in synthetic carbohydrate chemistry. To implement this capability, we introduced a jacketed reactor depicted in Figure 1B. The coolant was delivered by an external submersible pump. The cooling rate was controlled via relay transduction of the electrical signal from the rewired line D of HPLC Pump 1. Similarly to the previous set up, the reactor was also equipped with PLC that operates capacitive proximity sensor installed at the bottom drain of the reactor, an optical pathway sensor, and a solenoid valve. For a complete technical description of the components refer to Supporting Information.

To test low-temperature reaction capability, we performed glycosidation of thioglycoside donor **12**[36] with primary acceptor **13**[37] in the presence of *N*-iodosuccinimide (NIS)/TfOH promoter system in DCM/1,4-dioxane at −20 °C. 1,4-Dioxane was found to be helpful to solubilize NIS, which is poorly soluble in DCM. Complete solubility is necessary to avoid clogging tubing involved in the upstream operations. Conventional manual glycosylation of acceptor **13** with donor **12** afforded glycoside **14** in 70% yield (entry 3, Table 3). After some optimization of upstream operations and reaction conditions, the automated synthesis and separation afforded the desired glycoside **14** in 80% yield (entry 4).

We then investigated the glycosylation of sugar acceptor **1** with thioglycoside **15**.[38] A manual reaction in the presence of NIS/TfOH in DCM and MeCN at −20 °C produced disaccharide **16**[39] in 71% in 1.5 h (entry 5). MeCN co-solvent was found to be helpful to solubilize NIS and allowed to achieve the best results in this application. Analogous automated reaction afforded disaccharide **16** in 70% yield under same conditions (entry 6). Subsequently, we investigated glycosylation of cholesterol acceptor **18** with thioglycoside donor **17**[40] under similar reaction conditions. Both manual and automated syntheses were successful, and the desired glycolipid **19**[41] was obtained in 88% and 84%, respectively (entries 7–8). We note that the automated synthesis-purification yielded product **19** in 95% purity. Finally, we performed glycosylation of diol acceptor **20**[42] with donor **17**. This double glycosylation was nearly equally efficient under manual or automated set-up, and trisaccharide **21** was obtained in 76–79% yield (entries 9–10). These reactions were also conducted in the presence of NIS and TfOH at −10 °C. The automated synthesis took little longer (1.5 h vs 1 h), and the reagent ratios have remained practically the same as that in the manual approach.

Conclusions

As the 21st century unfolds with rapid changes, new challenges in research and development emerge. HPLC–A described herein

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uses standard HPLC components, a computer interface, and standard HPLC operating software. The delivery of standard ACS-spec solvents ensures authentication of key chemical resources and helps reproducibility of experiments. The key strategic advancements that sets this HPLC–A platform from other approaches are as follows:

1. Reactor equipped with the stirrer helps to solve the mixing problems that hamper all flow-based reactions
2. Using the reactor dramatically simplifies transitioning from regular manual reactions in the flask to the automated set-up. Although not strictly necessary, very minor adjustments to the ratio of reagents, and reaction time were helpful to enhance the reaction yields. In some applications, automated synthesis produced products in higher yields as compared to manual synthesis.
3. Both heating and cooling capabilities have been implemented, and the temperature control was achieved by using standard HPLC software via rewiring and interfacing the HPLC pump intake line.
4. The incorporation of an external programmable logic controller assisted by inexpensive proximity and optical sensors allowed to control the transfer process and loading the reaction mixture onto the column.
5. Promising results for separation of the produced reaction mixtures were achieved with the use of either disposable or recyclable home-packed flash chromatography columns.

With the fully automated platform we were able to record and save entire synthetic sequences as a computer program. The recorded sequences are all available as a part of the supporting information. These experiments can be accurately reproduced anywhere and by anybody who has access to HPLC, even non-specialists, who may not have expertise to do conventional synthesis. This work describes our first steps towards developing a HPLC-based automation platform that is applicable to the synthesis of building blocks, glycosylation, glycan assembly, and purification using the same universal set-up.

Supporting Information Summary

This material is available free of charge via the Internet.

Author Contributions

All authors have given approval to the final version of the manuscript.

Acknowledgements

The authors thank the National Institute of General Medical Sciences (GM111835, GM120673, GM152125) for the support of this work. L.P. was supported by EU funding within the NextGeneration EU-MUR PNRR PRIN 2022 (Prot. 20224LLK82).

Conflict of Interests

The authors declare no competing financial interests.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: Automation · Glycans · Synthesis · Glycosylation · HPLC

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Manuscript received: March 26, 2024
Accepted manuscript online: April 29, 2024
Version of record online: May 29, 2024

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